O-H-207B
August 16, 1977
SUPERSEDING
O-H-207a
March 12, 1964

FEDERAL SPECIFICATION

HERBICIDE, SIMAZINE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of pre-emergence herbicides known as simazine which are formulated with 2-chloro-4,6-bis-(ethylamino)-s-triazine, and inert ingredients, as either a water dispersible powder for application in water suspension as a spray, or as a dry granular product, for the nonselect control of vegetation (see 6.1).

1.2 Classification. The simazine herbicide shall be of the following types, as specified (see 6.2).

Type I - Water dispersible powder (80 percent concentration of 2-chloro-4,6-bis-(ethylamino)-s-triazine)

Type II - Granular product (4 percent concentration of 2-chloro-4,6-bis-(ethylamino)-s-triazine)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

RR-S-366	-	Sieve, Test
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter
PPP-D-723	-	Drums, Fiber

FSC 6840

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Federal Standards:

FED-STD-123 - Marking for Shipment (Civil Agencies)
FED-STD-791 - Lubricants, Liquid Fuels, and Related Products, Methods of Testing

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads for 40" X 48" Pallets

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

Federal Insecticide, Fungicide, and Rodenticide Act

(Title 7 United States Code Section 135 et seq.) as amended by the Federal Environmental Pesticide Control Act of 1972 (PL 92-516) and any regulations promulgated thereunder by the Administrator of the Environmental Protection Agency.)

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(Application for copies should be addressed to the Environmental Protection Agency, Pesticide Registration Division, 401 M Street, S.W., Washington, DC 20460.)

Title 49, Code of Federal Regulations (CFR), Parts 170 through 179, Hazardous Materials Regulations

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above should cite "49 CFR 170 to 179".)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N. W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

World Health Organization

Specifications for Pesticides Used in Public Health; Fourth Edition, 1973, Geneva

(Copies of the WHO Specifications for Pesticides may be obtained from the American Public Health Association, Inc., 10015 Eighteenth Street, N.W., Washington, DC 20036.)

3. REQUIREMENTS

3.1 Type I. Water dispersible powder.

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3.1.1 Composition. The simazine herbicide powder shall be composed of 2-chloro-4,6-bis-(ethylamino)-s-triazine and such inert diluting, modifying, and conditioning agents as are needed to meet the requirements of this specification. The concentration by weight of 2-chloro-4,6-bis-(ethylamino)-s-triazine in the simazine herbicide powder shall be not less than 80 percent, when tested as specified in 4.4.1.1. The powder shall contain a maximum of 2 percent by weight of material soluble in n-hexane when tested as specified in 4.4.1.2, and a maximum of 5 percent by weight of water when tested as specified in 4.4.1.3.

3.1.2 Chemical stability. The concentration of 2-chloro-4,6-bis-(ethylamino)-s-triazine required in 3.1.1 shall be maintained when tested as specified in 4.4.1.4.

3.1.3 Reaction. The acidity of the powder shall be not greater than 2 percent calculated as H_2SO_4 when tested as specified in 4.4.1.5.1. The alkalinity of the powder shall be not greater than 2 percent calculated as NaOH, when tested as specified in 4.4.1.5.2.

3.1.4 Particle size.

3.1.4.1 Average diameter. The surface-mean particle diameter of the powder shall be not greater than 3 microns (0.003 mm) when tested as specified in 4.4.1.6.1.

3.1.4.1 Maximum diameter (under simulated storage conditions conducive to caking). Not less than 98 percent of the powder shall pass through a 74-micron (0.074 mm) (U.S. No. 200) sieve, conforming to RR-S-366, when tested as specified in 4.4.1.6.2.

3.1.5 Dispersibility. Not less than 95 percent of the powder incorporated in suspension shall pass through a 74-micron (0.074 mm) (U.S. No. 200) sieve, conforming to RR-S-366, when tested as specified for Test A (without pretreatment) in 4.4.1.7.3.1; and not less than 90 percent shall pass through the same sieve specified herein, when tested as specified for Test B (with pretreatment) in 4.4.1.7.3.2.

3.1.6 Foaming. The powder shall form not more than 60 mm of foam instantaneously, and not more than 25 mm of foam after standing 5 minutes, when tested as specified in 4.4.1.8.

3.1.7 Appearance and pour. The powder shall be free-flowing, dry, loose, free from lumps and foreign matter, and move smoothly in an unimpeded stream when tested as specified in 4.4.1.9.

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3.2 Type II. Granular product.

3.2.1 Composition. The simazine herbicide granular product shall be composed of such active ingredients and inert ingredients as are specified herein. The product shall be prepared by impregnating and coating with a slurry of the active ingredient, and shall consist essentially of dry granular aggregates that can be broken down by weathering to further the distribution of the herbicide after application to the soil. Before being packaged, the final product shall be cured for a sufficient time to allow full absorption of the solution of the active ingredient.

3.2.1.1 Active ingredient. The active ingredient shall be 2-chloro-4,6-bis-(ethylamino)-s-triazine and it shall be in a concentration by weight of not less than 4 percent when tested as specified in 4.4.2.1.

3.2.1.2 Inert ingredients. The inert ingredients shall consist of (1) the granular carrier, (2) the unevaporated liquid hydrocarbon remaining permanently as part of the dry finished preparation and (3) any impurities in the active ingredient used in preparation of the granular herbicide.

3.2.1.2.1 Carrier. The carrier shall be a granular attapulgite clay or other mineral that has the ability to hold and carry relatively large amounts of oily, waxy, or sticky chemicals while retaining a dry and free-flowing condition in the finished product. It shall be extruded (commonly designated AA) and uncalcined (commonly designated Regular or RVM). It shall produce a stable formulation with the active ingredient either with or without a deactivator, and also have sufficient absorptive capacity to inhibit agglomeration, caking, and lumping.

3.2.1.2.2 Solvent. The liquid hydrocarbon used in making the solution of the active ingredient for impregnation of the granules shall be a suitable hydrocarbon, having the following characteristics, as determined in accordance with FED-STD-791:

		Test method
Flash point, minimum °F/°C. (Cleveland Open Cup).....	140/60	1103.8
Distillation range, °F/°C:		1001.13
Initial boiling point, minimum.....	320/160	
50 percent point, minimum.....	400/204	
End point, maximum.....	660/345	
Specific gravity, 60°F/60°F. (15.6°C/15.6°C)	0.890 to 0.960	401.7
Viscosity, SSU at 100°F (37.8°C) maximum.....	40 sec.	304.8

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3.2.2 Chemical stability. Not more than 10 percent of the concentration of 2-chloro-4,6-bis-(ethylamino)-s-triazine specified in 3.2.1.1 shall be lost from the product when tested as specified in 4.4.2.2.

3.2.3 Reaction. The acidity of the product shall be not greater than 2 percent calculated as H_2SO_4 and the alkalinity shall be not greater than 2 percent calculated as NaOH, when tested as specified in 4.4.2.3.

3.2.4 Grain size distribution. The grain size distribution of the product shall meet the requirements in table I. The distribution shall be determined by the dry sieve test specified in 4.4.2.4.

TABLE I. Grain size distribution (dry sieve)

Sieve		Percent passing	
Micron size	U. S. No.	Minimum	Maximum
2000 (2 mm)	10	99.8	-
590 (0.59 mm)	30	-	10
250 (0.25 mm)	60	-	1.5

3.2.5 Grain breakdown. Not less than 75 percent of the product shall pass a 250-micron (0.25 mm) (U. S. No. 60) sieve when tested as specified in 4.4.2.5.

3.2.6 Appearance and pour. The granular product shall be free-flowing, dry, loose, non-dusty, non-gritty, free from lumps and foreign matter, and move smoothly in an unimpeded stream when tested as specified in 4.4.2.6.

3.3 Labeling. The herbicide shall be registered with the Pesticide Registration Division, Environmental Protection Agency, and shall be in compliance with the requirements of the Federal Insecticide, Fungicide and Rodenticide Act, as amended, of 1972.

3.3.1 Label data. Label data shall be durably and legibly marked on the main body, by lithographing or stenciling with non-corrosive, weather-resistant, waterproof ink which will not smear or rub off during handling. Label data shall include, but not be limited to the following:

(a) The name and percent by weight of each active ingredient.
(b) The total percent by weight of inactive ingredients.
(c) National stock number, Federal specification number, contract number, and lot number.
(d) Manufacturer's name and address.
(e) EPA registration and establishment number.

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(f) Directions for use: As a minimum, the label shall list useage directions for the nonselective control of vegetation and examples of species of plants resistant to the herbicide or difficult to kill.

(g) Precautionary information: CAUTION. KEEP OUT OF REACH OF CHILDREN. Harmful if swallowed. Avoid breathing dust, spray mist or vapors. Handle in a ventilated area. Avoid contact with skin, eyes or clothing. In case of contact immediately flush eyes or skin with plenty of water. Get medical attention if irritation persists. Wash hands thoroughly after handling. Wash contaminated clothing before reuse. Avoid contamination of food and feed. Do not contaminate domestic or irrigation water supplies. Do not contaminate water by cleaning of equipment or disposal of wastes. Apply this product only as specified on this label. Do not reuse container. Destroy when empty. Store in a dry place.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated herein.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended or qualified in this specification or applicable purchase document.

4.3 Inspection of the end item.

4.3.1 Examination of the end item. The end item shall be examined for the specific defects in 4.3.1.1 and 4.3.1.2. The applicable inspection levels and acceptable quality levels (AQLs) shall be as indicated in 4.3.2. The lot size for purposes of determining the sample size shall be in accordance with MIL-STD-105. The sample size shall be expressed in filled unit containers of the same capacity.

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4.3.1.1 Examination of unit container for visual defects. The sample unit for this examination shall be one filled unit container.

Examine	Defect
Unit container for:	
Workmanship	Dented, abraded, dirty, split, punctured or any seepage of contents. Bulged or distorted.
Marking or labeling	Missing, incomplete, incorrect, illegible.
Contents	Not homogenous, contains lumps, evidence of moisture, presence of foreign matter.

4.3.1.2 Examination of filled container for net contents. The sample unit for this examination shall be one filled unit container. The lot shall be unacceptable if the average net contents per container for all sample units examined is less than specified or indicated.

4.3.2 Inspection levels and acceptable quality levels (AQLs). The inspection levels and AQLs expressed in defects per 100 units, shall be as follows:

Examination	Inspection level	AQL
4.3.1.1	I	2.5
4.3.1.2	S-2	N/A

4.3.3 Examination of preparation for delivery. Inspection for packaging, packing and marking shall be performed in accordance with the quality assurance provisions contained in PPP-D-723 or the appendix of PPP-C-96, as applicable.

4.3.4 Examination for palletization. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully prepared for delivery. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL shall be 6.5 defects per 100 units.

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Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirement.
Palletization	Not as specified. Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.

4.3.5 Testing of the end item. The end item shall be tested for the characteristics specified in table II. The lot size shall be expressed in pounds (0.45 kilograms) of herbicide. The sample unit for testing shall be a 1-pound (0.45 kilograms) composite obtained by combining and thoroughly mixing equal portions from containers selected at random throughout the lot. The composite shall be placed in a clean, dry, amber glass container and sealed. Care shall be exercised to prevent contamination or alteration of the herbicide during the sampling, compositing, storage and testing. All test reports shall contain the individual values used in expressing the final result. The lot shall be unacceptable if the composite fails to meet any test requirements specified. The lot size and sample size shall be in accordance with the following:

Lot size (pounds/kilograms)	Sample size
800/362.9 or less	2
801/363.3 up to and including 22,000/9979	3
22,001/9979.5 and over	5

TABLE II. Testing of the end item

Characteristic	Requirement paragraph	Test method	Results reported as Pass/fail	Numerically to nearest
Simazine concentration	3.1.1, 3.2.1.1	4.4.1.1		1 percent
n-Hexane solubles	3.1.1	4.4.1.2		1 percent
Water content	3.1.1	4.4.1.3		1 percent
Chemical stability	3.1.2, 3.2.2	4.4.1.4		1 percent
Reaction	3.1.3, 3.2.3	4.4.1.5		1 percent
Particle size	3.1.4	4.4.1.6		1 percent
Dispersibility	3.1.5	4.4.1.7		1 percent
Foaming	3.1.6	4.4.1.8		1 mm
Appearance and pour	3.1.7, 3.2.6	4.4.1.9	X	
Grain size distribution	3.2.4	4.4.2.4		1 percent
Grain breakdown	3.2.5	4.4.2.5		1 percent

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4.4 Test methods.

4.4.1 Type I, water dispersible powder.

4.4.1.1 Concentration of 2-chloro-4,6-bis-(ethylamino)-s-triazine. The concentration is calculated from the organically bound chlorine content. The data required to make the calculations are obtained from the determination of the total organic chlorine content and the ionic chlorine content. This method is not applicable if other compounds containing organic chlorine are present.

4.4.1.1.1 Reagents.

(a) Morpholine, commercial grade.
(b) 50 percent Sulfuric Acid (H_2SO_4).
(c) 0.1N Silver Nitrate, standardized ($AgNO_3$).
(d) Dimethylformamide (DMF), technical grade.
(e) pHydrion paper, 6.0 to 8.0 pH range.

4.4.1.1.2 Apparatus.

(a) 100-ml round bottom flask with 24/40 ground glass opening.
(b) Water cooled reflux condenser with a 24/40 ground glass fitting.
(c) Water bath heated by steam coils.
(d) 50-ml burette.
(e) Beckman Model H-2 pH meter, or equivalent.
(f) Electrode, silver and calomel.
(g) Magnetic stirrer with teflon stirring bar.

4.4.1.1.3 Procedure.

(1) Total organic chlorine:

Weigh to the nearest milligram, a 0.94-gram sample (W_1) into the round bottom flask. Add 20 ml of morpholine. Connect the condenser to the flask and place on a boiling water bath for 15 to 30 minutes. Rinse contents of flask into a 400-ml beaker with distilled water. Acidify slightly with 50 percent H_2SO_4 using pHydrion paper. Cool to room temperature and titrate potentiometrically with standard $AgNO_3$ solution. The milliliters of standard $AgNO_3$ required = a.

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(2) Ionic Chlorine:

Weigh to the nearest milligram, 0.94-gram sample (W_2) into a 400-ml beaker. Add 20 ml of dimethylformamide and boil for 5 minutes on the water bath. Dilute with 300 ml of distilled water and acidify slightly with 50 percent H_2SO_4 using pHydrion paper. Cool to room temperature. Titrate potentiometrically with standard $AgNO_3$ solution. The milliliters of standard $AgNO_3$ required = b.

Calculations:

$$\frac{(a-c)N \times 201.66}{W_1 \times 1000} \times 100 = \text{Percent 2-chloro-4,6-bis-(ethylamino)-s-triazine}$$

Where:

a = ml of $AgNO_3$ in organic chlorine determination

b = ml of $AgNO_3$ in ionic chlorine determination

$c = b \times \frac{W_1}{W_2}$

W_1 = weight of sample used in organic chlorine determination

W_2 = weight of sample used in ionic chlorine determination

N = Normality of standard $AgNO_3$

4.4.1.2 Determination of material soluble in n-hexane. Weigh 3 $\pm$ 0.1 grams of the herbicide sample into a 200-ml Erlenmeyer flask with ground-in reflux condenser. Add 100 ml of n-hexane and reflux for 15 minutes. Cool to 10°C with frequent stirring, filter through a sintered-glass crucible, and wash with n-hexane at 0°C. Rinse the filtrate into a weighed Erlenmeyer flask and distill off the solvent. Dry at 100°C for 30 minutes, cool, and weigh.

4.4.1.3 Water content. The test method to be used for the determination of water content of the herbicide shall be that contained in the 1973 edition of the World Health Organization (WHO) Specifications for Pesticides used in Public Health. (Karl Fischer electrometric titration method).

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4.4.1.4 Chemical stability. This test is intended for the determination of any chemical breakdown of the herbicide when stored at a high temperature.

4.4.1.4.1 Apparatus.

(a) Oven, regulated to maintain a temperature of 55° ± 2°C.

(b) Storage jars: glass, 60 to 120 ml with screw-caps and aluminum inserts in caps.

4.4.1.4.2 Procedure. Load a clean, dry storage jar approximately half full of the herbicide sample without compacting. Close the jar tightly with the aluminum insert cap, and store in the oven at 55° ± 2°C for 14 days.

Remove the jar from the oven and allow to cool to room temperature. Weigh out a 1.0-gram sample, and analyze for 2-chloro-4,6-bis-(ethylamino)-s-triazine as described in 4.4.1. Calculate the loss, if any, in terms of percent of 2-chloro-4,6-bis-(ethylamino)-s-triazine originally present.

4.4.1.5 Reaction.

4.4.1.5.1 Acidity.

4.4.1.5.1.1 Procedure. Weigh 10 ± 0.1 grams of the herbicide sample and disperse in 100 ml of distilled water, stirring frequently for 1/2 hour. Filter and titrate the filtrate immediately with 0.02N sodium hydroxide (NaOH), using methyl red as indicator. Carry out a blank determination on 100 ml distilled water with 0.02N NaOH.

Calculation:

Acidity as H_2SO_4 (percent) = 0.0098 X (a-b)

where a = volume in ml of 0.02N NaOH used for the sample

b = volume in ml of 0.02N NaOH used for the blank

NOTE: The blank may take the form of a small titre with 0.02N HCl, in which case

Acidity as H_2SO_4 (percent) = 0.0098 X (a + c)

where a = volume in ml of 0.02N NaOH used for the sample

c = volume in ml of 0.02N HCl used for the blank

Alternatively, the end-point may be determined electrometrically.

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4.4.1.5.2 Alkalinity.

4.4.1.5.2.1 Procedure. Weigh 10 ± 0.1 g of the herbicide sample and disperse in 100 ml of distilled water, stirring frequently for 1/2 hour. Filter, and titrate the filtrate immediately with 0.02N hydrochloric acid (HCl), using methyl red as indicator. Carry out a blank determination on 100 ml of distilled water with 0.02N NaOH.

Calculation:

Alkalinity as NaOH (percent) = 0.008 X (d + e)

where d = volume in ml of 0.02N HCl used for the sample

e = volume in ml of 0.02N NaOH used for the blank

NOTE: The blank may take the form of a small titre with 0.02N HCl, in which case

Alkalinity as NaOH (percent) = 0.008 X (d - f)

where d = volume in ml of 0.02N HCl used for the sample

f = volume in ml of 0.02N HCl used for the blank

Alternatively, the end-point may be determined electrometrically.

4.4.1.6 Particle size.

4.4.1.6.1 Average diameter. The surface-mean particle diameter shall be determined by the air-permeation method. The apparatus used shall be a Government approved air-permeation apparatus (see 6.4). The solid density of the herbicide powder, knowledge of which is prerequisite to particle fineness measurement by the air-permeation procedure, may be determined by any standard method. A suitable method is the conventional method of liquid immersion in a specific gravity bottle having a ground-in thermometer, with distilled water as the immersion liquid. In the use of this method, care must be taken to guard against insignificant loss of water-soluble material; the pycnometer (specific gravity bottle) should have a nominal volume of 25 ml at least, and the weight of sample used for observation should be as great as can be handled conveniently in the pycnometer used.

4.4.1.6.2 Maximum diameter.

4.4.1.6.2.1 Apparatus.

(a) Pretreatment jar; an air-tight screw-cap glass jar, 35 ml ± 5 ml actual capacity.

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(b) Oven; regulated to maintain 55° ± 1°C.

(c) Mixing jar: a screw-cap glass jar, 295 to 300 ml (10 U.S. fl oz), substantially cylindrical, approximately 8 cm deep, without abrupt shoulders. (Some kinds of conventional fruit jars qualify.)

(d) Tumbling machine; any mechanical device for rotating the mixing jar around its center, end-over-end, at 60 r.p.m.

(e) Sieve; a 74-micron (0.074 mm) (U.S. No. 200) standard testing sieve, 8-inch (20 cm) frame diameter, conforming to RR-S-366.

(f) Sprayer; consisting of a shower head, with not less than 50 perforations distributed practically uniformly over a circular face-plate area 6 to 8 cm in diameter, connected by a hose to a source of tap water. The relation of water pressure to size and number of perforations must be such that if the shower head be inverted to send a spray upward no considerable portion of the spray will rise more than 15 cm above the face plate. If the pressure as limited by this criterion is insufficient to give the prescribed flow rate (see paragraph 3 below), the flow rate must be adjusted by increasing the size and/or number of perforations.

4.4.1.6.2.2 Procedure. This test, which is primarily a sieve test designed so that essentially all the material in an acceptable sample will pass through the sieve, involves a pretreatment to simulate storage conditions conducive to caking. The complete test procedure consists of three steps: (1) the pretreatment; (2) the preparation of a suspension of the treated sample in water; and (3) the actual screening and determining the percentage of residue. The three steps shall be performed in the following manner:

(1) Pretreatment. Store the opened jar and screw-cap at 55°C ± 1°C for at least a 1-hour period; then, without allowing the jar to cool, pour an 8-gram sample of the powder into the jar, seal the jar air-tight and keep it in the oven at 55°C ± 1°C for 24 hours. Remove from oven and let stand at room temperature, still sealed for 3 hours; then proceed promptly to preparation of suspension as follows:

(2) Preparation of suspension. (a) Deposit the 8-gram sample of pre-treated powder on a 50-ml portion of distilled water in the mixing jar and let stand for 3 minutes for the powder to sink and/or soak undisturbed. (b) Close the jar and tumble it on the machine at 60 r.p.m. for 5 minutes. (c) Dilute with 185 ml of distilled water. (d) Tumble as before, for 1 minute, and proceed as follows:

(3) Screening. Transfer the suspension to the sieve, which in preparation for the test must have been freed of any film of grease or other water-repellent material. Wash the residue on the sieve for 10 minutes with the sprayer adjusted to deliver 4 to 5 liters per minute. Transfer the residue to a filter whose weight has been determined on an oven-dried basis (oven temperature not over 80°C); dry the residue in an oven at 55°C ± 1°C and calculate the weight percentage of sample passing through the sieve.

4.4.1.7 Dispersibility.

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4.4.1.7.1 Reagents. Hard water; designed to have a hardness of 342 p.p.m. calculated as calcium carbonate; composition as follows:

Calcium chloride anhydrous.....	0.304 gram
Magnesium chloride hexhydrate.....	0.139 gram
Distilled water to make.....	1 liter of solution

4.4.1.7.2 Apparatus.

Same as for maximum diameter test (see 4.4.1.6.2) with omission of sprayer.

4.4.1.7.3 Procedure. The dispersibility of the herbicide powder in water is tested by the following special sieve tests in which the suspension is simply poured upon the sieve without any washing or any agitation of the sieve. In preparation for these tests make sure that the wire cloth of the sieve is free from any film of grease or other water-repellent material and is throughly wetted with water on both sides and kept so wetted until the moment for use.

4.4.1.7.3.1 Test A: Without treatment. Prepare a suspension as specified in subparagraph (2) of 4.4.1.6.2.2 except to omit the pretreatment of the powder and to substitute hard water for distilled water throughout. When the final tumbling is completed, remove the jar from the machine, and with the sieve standing in a level position, pour the suspension as evenly as possible over its whole area, taking care meanwhile not to allow any powder to settle in the jar. Raise one side of the sieve frame 25 mm by means of a prop and allow the residue to drain undisturbed for 5 minutes. Transfer the residue to a filter, dry and weigh as in subparagraph (3) of 4.4.1.6.2.2 and calculate the dispersibility at 74 microns (0.074 mm) as weight percentage of sample passing through sieve. (Filtration may often be facilitated by washing the residue from the sieve into a beaker, adding a few ml dilute alum solution to flocculate the solids, then pouring onto a coarse filter.)

4.4.1.7.3.2 Test B: With pretreatment. The procedure is the same as for test A, except that the pretreatment specified in subparagraph (1) of 4.4.1.6.2.2 is included.

4.4.1.8 Foaming.

4.4.1.8.1 Apparatus. The apparatus is shown in figure 1 and is made to standard dimensions from stock glass. The apparatus consists of a reservoir or pipette (A), a receiver (B), a jacket (C), and a supporting stand and other parts required to make a complete assembly. The orifice of the reservoir is so constructed that the turbulence set up cause the stream of solution to break up into droplets just after emerging from the orifice. Foam is produced when these droplets strike the surface of the liquid or foam in the receiver.

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Construct the tip of the reservoir from pyrex glass tubing whose outside diameter is approximately 7 mm, and length is 6 cm. Select another piece of tubing whose outside diameter is such that it fits snugly inside the 7 mm piece. From this cut a short 1-cm section and grind it square at both ends. Place this inner section flush with the extreme tip of the large piece and then heat the outside of the larger tube to redness at one or two points opposite the middle of the inner section. Join the tip assembly to the reservoir proper.

Orifices varying in size from 2 mm to 3 mm in diameter produce nearly the same amount of foam. In standardizing several pieces of apparatus to conform to each other, it is permissible to alter slightly the solution volume in the reservoir.

4.4.1.8.2 Procedure. Prepare a suspension in the ratio of 45.36 g active ingredient in 1 gallon (3.8 liters) of hard water (see 4.4.1.7.1) at 18° to 24°C.

To set up and operate the test, proceed as follows:

(1) Adjust the receiver so that it is plumb.
(2) Adjust the reservoir so that a stream of water emerging from the orifice will strike the bottom center of the receiver.
(3) Be sure that both the receiver and reservoir are clean. The final step in the cleansing shall be a thorough rinsing with distilled water. If the glass surfaces are clean, the distilled water should flow over the side walls in an unbroken film, finally draining without the appearance of droplets on the glass.
(4) Circulation shall be maintained through the jacket at a controlled temperature of 18° to 24°C.
(5) Close the stopcock of the receiver and by means of a 50-ml volumetric pipette introduce 50-ml of the test suspension into the receiver. The suspension should be run down the side walls of the receiver to avoid foam formation and to wet the entire interior of the receiver. Hold the tip of the pipette against the side wall of the receiver and slowly move the tip around the circumference of the tube so that the suspension will flow down on all sides.
(6) Fill the reservoir in the manner of filling a volumetric pipette. The reservoir should be marked at a point which indicates a liquid content of 200 ml.
(7) Place the reservoir in position at the head of the receiver and open the reservoir stopcock.
(8) When all the liquid has run out of the reservoir, start stopwatch and take foam height readings immediately and after 5 minutes. The first reading is the accepted standard but additional indications of the relative stability of various foams may be obtained in those cases where breakdown does occur in 5 minutes.

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(9) The readings should be taken as follows: A milliliter scale is placed behind the receiver so that the zero mark is opposite the point in the receiver which would be reached by the total volume of liquid introduced into the receiver. The foam production is measured at the top of the foam column at the highest average height to which the rim of the foam has reached. This height is proportional to the volume of air remaining in the foam. Additional measurements of foam height may be taken at suitable intervals.

4.4.1.9 Appearance and pour. Measure 50 ml of dry herbicide sample into a glass cylinder 7 inches (17.8 cm) high by 7/8 inch (2.2 cm) in diameter, and equipped with a spout. By tilting the cylinder pour the herbicide at a uniform rate of approximately 100 ml per minute. Observe the appearance of the herbicide in the cylinder and the pouring for conformance to 3.1.7.

4.4.2 Type II, granular product.

4.4.2.1 Concentration of 2-chloro-4,6-bis-(ethylamino)-s-triazine. The concentration shall be determined by the test specified in 4.4.1.1, with the exception that the weight of each sample shall be 18.75 g instead of 0.94 g.

4.4.2.2 Chemical stability. The chemical stability shall be determined by the test specified in 4.4.1.4.

4.4.2.3 Reaction. The reaction shall be determined by the tests specified in 4.4.1.5.

4.4.2.4 Grain size distribution.

4.4.2.4.1 Sieves. Sieves used in this test shall conform to RR-S-366. Mesh sizes are 2000-micron (2 mm) (U.S. No. 10), 590-micron (0.59 mm) (U.S. No. 30), and 250-micron (0.25 mm) (U.S. No. 60). Sieve frame is 8-inch (20 cm) diameter either full height (inside depth 2 inches (5 cm)) or half height (inside depth 1 inch (2.5 cm)).

4.4.2.4.2 Sample preparation. Each time any portion is removed from the stock sample, extreme care must be taken to include the proportionate amount of dust with the granules, in order that both the with-drawn sample and the remaining stock be representative in dust content. This necessity is the basis for the following procedure, to be followed in every instance.

(a) Determine the weight of the stock sample.
(b) Tumble the stock sample end-over-end at approximately 30 r.p.m. for 1 minute in a jar whose volume is about twice the apparent volume of the sample.

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(c) Separate a 20- to 25-gram (± 0.1 g) portion as a test sample by turntable sampling as follows: A sieve receiver pan, 8-inch (20 cm) diameter, 2-inch (5 cm) depth, is secured in a centered position on a 45 r.p.m. turntable (see 6.5). It is recommended that the drive mechanism for any speeds other than 45 r.p.m. be inactivated to prevent their inadvertent use. A sheet-metal receiver cup, with plain, thin unflanged rim is placed within the pan. A 60° glass funnel is adjusted on a support so that the outlet tip just clears the top of the cup, and the funnel and cup are set off-center with respect to the turntable by such an amount that the arc projected by the funnel tip across the cup in rotation of the table will bear the same ratio to the complete circle as the weight of the desired test sample bears to the weight of the whole stock sample. The bore diameter of the funnel stem should be just sufficient to provide free flow for the granular material being treated. With the turntable rotating at 45 r.p.m. the entire stock sample is poured through the funnel so as to be delivered continuously in a steady stream. In case of clogging or bridging, normal flow is restored (the table still turning) by probing with a stout wire flattened on the lower end. The portion for test falls in the cup, while the remainder falls in the pan outside the cup. The cup is removed, any dust clinging to the outside surfaces is brushed back into the pan, and the cup with its load is weighed for determination of the test sample weight; the remainder in the pan is returned to the stock jar.

4.4.2.4.3 Procedure. Screen the test sample through a nest of sieves of the designated mesh sizes, using a single-eccentric type mechanical shaker that imparts to the sieve a rotary motion and tapping action of uniform speed of approximately 300 gyrations and approximately 150 taps per minute. Continue the screening for 15 minutes. Weigh the residues and calculate the percentage passing through each sieve. In the weighing and transferring of the sieve sample and its size fractions, any dust clinging to sieve cloth, sieve wall, scale pan and miscellaneous surfaces, or clogging the sieve-cloth apertures, is to be regarded as part of the finest fraction (sieve-pan fraction). Neither this dust nor the material actually entering the sieve pan need be recovered for weighing; the granules lying on the respective sieve cloths are to be recovered and weighed, and the pan fraction calculated by difference.

4.4.2.5 Grain breakdown.

4.4.2.5.1 Sieve. The sieve used in this test shall conform to RR-S-366. The mesh size is 250-micron (0.25 mm) (U.S. No. 60), and the sieve frame is 8-inch (20 cm) diameter, full height (inside depth 2 inches (5 cm)).

4.4.2.5.2 Sample preparation. Sample preparation shall be the same as that specified in 4.4.2.4.2 except that the sample weight in paragraph (c) shall be 10- to 12 g instead of 20- to 25 g.

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4.4.2.5.3 Procedure. Transfer the test sample to a 250 ml glass-stoppered graduated cylinder of conventional design and make up to 250 ml volume with distilled water at a room temperature of 25°C ± 1°C. Rotate the cylinder end-over-end in a tumbling machine for 2 hours at 30 r.p.m. Remove the cylinder, allow it to stand undisturbed for 30 minutes, and draw off by suction and discard any oil that has risen to the top. Support the cylinder upright in a clamp over the sieve so that the base of the cylinder is approximately centered within the sieve frame and is approximately 1 cm above the sieve cloth. Flow the sample onto the sieve by means of a vigorous stream of tap water introduced through a hose extending down into the cylinder nearly to the bottom. If the last few grains persist in the cylinder, remove them by inversion of the cylinder, using water but not the hose. As soon as the entire sample has been washed from the cylinder, transfer the sieve residue to a filter by rinsing with a gentle stream of water. Dry the residue at a temperature of 50°C ± 5°C to constant weight and calculate the percent of sample passing through the sieve.

4.4.2.6 Appearance and pour. This test is the same as that specified in 4.4.1.9. Observe the appearance of the herbicide in the cylinder and the pouring for conformance to 3.2.6.

5. PREPARATION FOR DELIVERY

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. The herbicide shall be packaged in 1-pound (.45 kg) or 5-pound (2.3 kg) quantities in metal cans conforming to type V, class 2 of PPP-C-96, or in 50-pound (22.7 kg) quantity in fiber drums conforming to type III, grade A, class 2 of PPP-D-723 as specified (see 6.2).

5.1.2 Level C. The herbicide shall be packaged to afford adequate protection against physical damage during shipment from the supplier to the first receiving activity. The package shall be the same as that normally used by the supplier for retail distribution. The quantity per package shall be as specified (see 6.2).

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. The herbicide, packaged in metal cans as specified in 5.1.1, shall be packed for level A shipment in accordance with the requirements specified in the appendix of PPP-C-96. The herbicide, packaged in fiber drums as specified in 5.1.1, shall not require overpacking.

5.2.2 Level B. The herbicide, packaged in metal cans as specified in 5.1.1, shall be packed for level B shipment in accordance with the requirements specified in the appendix of PPP-C-96. The herbicide, packaged in fiber drums as specified in 5.1.1, shall not require overpacking.

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5.2.3 Level C. The herbicide, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the supplier for retail distribution. Container shall comply with the Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Palletization.

5.3.1 Metal cans. When specified (see 6.2), the herbicide in metal cans, packed as specified in 5.2, shall be palletized in accordance with the requirements specified in the appendix of PPP-C-96.

5.3.2 Fiber drums. When specified (see 6.2), the herbicide, packed in fiber drums as specified in 5.2, shall be palletized in accordance with load type XIII of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking.

5.4.1 Civil agencies. In addition to the marking specified in 3.3 and any special marking required by the contract or order, shipments shall be marked in accordance with FED-STD-123 and in accordance with the marking and labeling requirements of Title 49, Code of Federal Regulations, Parts 170 through 179, Hazardous Materials Regulations.

5.4.2 Military requirements. In addition to the marking specified in 3.3 and any special marking required by the contract or order, shipments shall be marked in accordance with MIL-STD-129 and in accordance with the marking and labeling requirements of Title 49, Code of Federal Regulations, Parts 170 through 179, Hazardous Materials Regulations.

6. NOTES

6.1 Intended use. The pre-emergence simazine herbicides covered by this specification are intended for use in the nonselective control of vegetation on non-cropland in storage yards and parking lots, on railroad beds and ballast, under pipelines, under guide rails and surrounding sign posts on highways, on public utility right-of-ways, around buildings and tank farms, along ditchbanks and fence rows, around antenna installations and runway lights, on crushed rock blankets, and in similar areas where any plant growth is undesirable. The granular type is particularly useful for soil applications where a general foliage deposit is not required or may be undesirable, in inaccessible places, or where water is in limited amounts.

Since simazine presents no electrical or conductivity problems, it is especially suitable around utilities, power plants, transformers, signal blocks, or other electrical applications. Since it is almost inactive on foliage there is no drift hazard.

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Simazine is carried into the soil by rainfall and acts through the root system. As germination takes place the vegetation is killed when it starts to emerge. It is practically insoluble in water and has a relatively long residual action.

Although simazine is a nonselective herbicide, there are certain species of plants resistant to it or difficult to kill. Among the resistant species are buttercup (creeping), dewberry, false golden aster, nutgrass, and vasey grass.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of simazine herbicide required (see 1.2).
(c) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(d) Size container required (see 5.1.1 and 5.1.2).
(e) When palletization is required (see 5.3).

6.3 Simazine is a common name accepted for use by the American Standards Association Sectional Committees on Common Names for Pest Control Chemicals K62 (ASA-K62).

6.4 Air permeation apparatus. An approved apparatus for use in conducting the particle size test in 4.4.1.6.1 is the self calculating air-permeation apparatus of the No. 2,261,802), or equal. The Sub-Sieve Sizer, manufactured by the Fisher Scientific Company, may be used as the test instrument. Each individual apparatus is subject to inspection by the agency procuring herbicide in order to verify calibration and performance.

6.5 Turntable. An approved turntable for use in conducting the grain size distribution test in 4.4.2.4.2 is the General Industries, Model TR, motorized 9-inch (23 cm) phonograph turntable, or equivalent.

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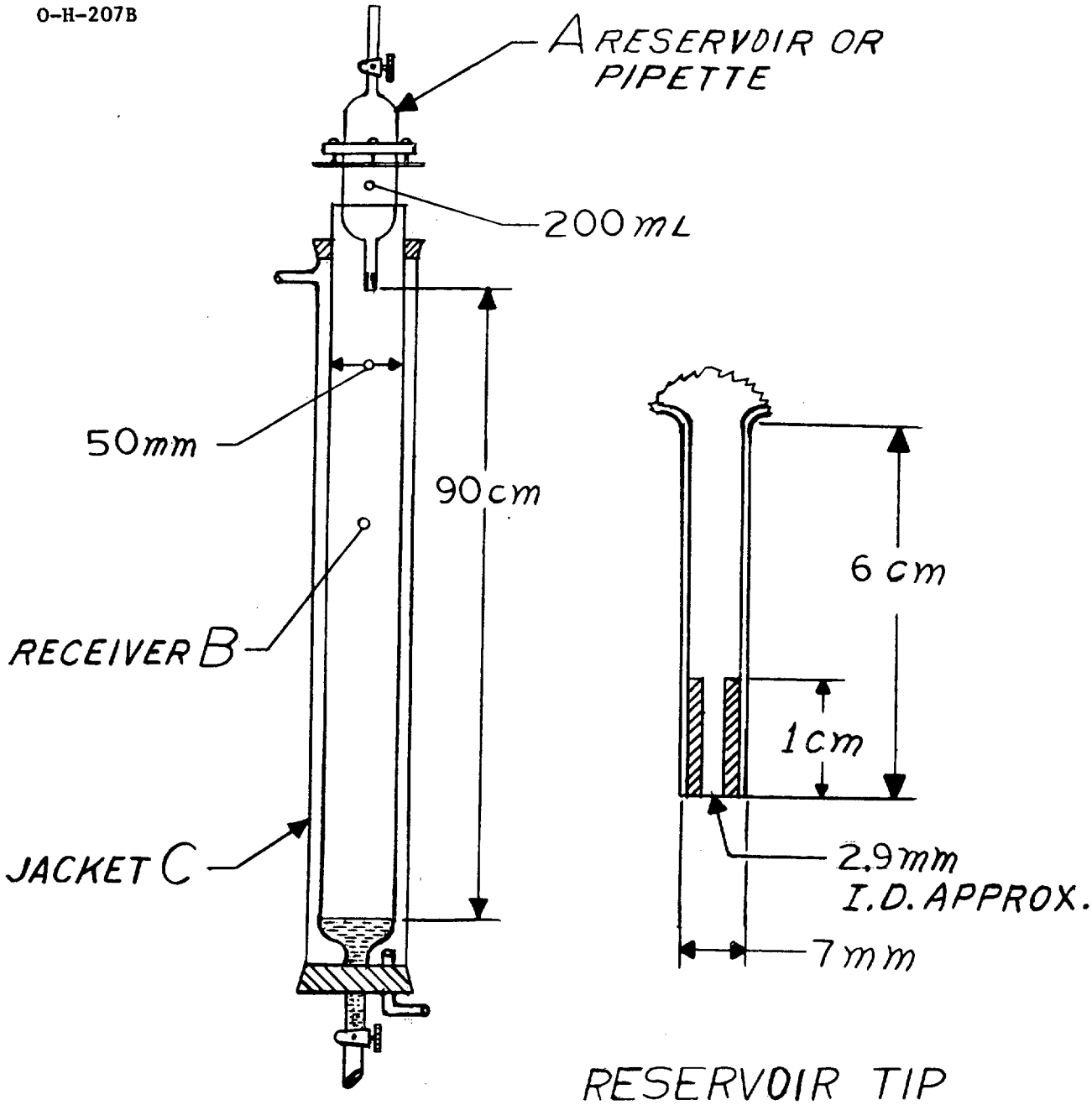


FIG 1 - FOAMING TEST APPARATUS

(SHOT AT 93%)

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